ROMEO DAVID JR.

Senior Telecommunications Engineer | Data & Automation Team Lead
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PROFESSIONAL SUMMARY

Senior Telecommunications Engineer with 19+ years of experience in multi-vendor network operations, project delivery, and operational automation. Currently leading O&M for IDEMIA MestaFusion ATMS across 100+ sites in KSA Central Region, achieving 97.8% YTD SLA compliance with an average MTTR of 21.2 hours. Independently developed practical Python and PowerShell automation tools and Power BI dashboards that reduce daily data-processing time by 65–80%. Combines deep telecom expertise with Generative AI and Agentic AI-assisted development to deliver efficient, real-world solutions.

CORE COMPETENCIES

Telecom Networks: TETRA • GSM/UMTS/LTE • Microwave Transmission (Multi-vendor)
Automation & Analytics: Python • PowerShell • Power BI Dashboards • Generative AI & Agentic AI-assisted Development
Operations Excellence: Site Surveys & Audits • Firmware Testing • SLA Management • Cross-regional Coordination
Leadership: Team & Sector Lead • Project Supervision • Stakeholder Communication

PROFESSIONAL EXPERIENCE

ASTEK Saudi Arabia Limited April 2021 – Present
Senior Telecom Engineer – Data & Automation Team Lead | Riyadh, KSA

- Lead O&M for IDEMIA MestaFusion ATMS across KSA Central Region; oversee technical support and cross-regional collaboration for 100+ live traffic monitoring sites.
- Achieved 97.8% YTD SLA compliance across 406 O&M tickets with zero overdue tickets and 100% restoration rate; maintained average MTTR of 21.2 hours.
- Independently developed and actively use six custom automation tools that transformed daily ATMS operations:
  - **Traffic Data Analysis Tool** – Processes traffic event XML files into daily/master CSV reports with geospatial analysis.
  - **Traffic Violation Manager (TVM)** – Simplifies case review by organizing violation files and media by site, type, and lane, with photo and video playback, EXIF verification, and Excel and one-page PDF reports.
  - **Hardware Serial Manager (HSM)** – Scrapes cabinet serials into the inventory for easier tracking, then syncs the confirmed update to Drive without duplicate rows.
  - **SMART Drive Health Monitor** – PowerShell-based field tool for remote SMART health monitoring of HDDs/SSDs via SSH; provides health recommendations and generates XML reports.
  - **Spare Usage** – Tracks spare-parts usage by site for restoration, PMR, and special tasks, with filters by spare type and monthly usage reports.
  - **Fleet Map** – Built a live map of the ATMS sites inside TicketFlow that works across different devices, with color-coded pins by ticket status, site thumbnails, and on-map weather with a 6-hour peek.
- Built Power BI dashboards visualizing ATMS traffic data, hourly trends, and site performance — enabling 65–80% faster reporting and decision-making.
- Led site surveys, audits, new site installations, and pre-rollout firmware/patch testing (providing feedback to IDEMIA France).

Houssam Contracting Company (HCC) April 2015 – April 2021
Telecom Engineer / Transmission Specialist / Sector Lead | Riyadh, KSA
Joint Venture of AIRBUS Defence and Space and Saudi Technical Trading Cont. Co.

- Led the Riyadh Sector for O&M of the TETRA-based Public Safety Radio Network (PSRN); managed team activities and delivered consolidated reports to NMC and Regional Manager.
- Supervised network upgrades, site relocations, and new TETRA site deployments from planning through installation, integration, and testing.
- Coordinated team resources to restore equipment within SLA and conducted site surveys and audits to maintain network integrity.

SMART Communications, Inc. July 2009 – February 2015
Senior Engineer | Philippines

- Executed end-to-end installation, commissioning, integration, and acceptance testing of multi-vendor transmission and cellular equipment (Nokia, Siemens, Ericsson, Aviat/Harris, Alcatel-Lucent, Huawei, Fujitsu, and others).
- Diagnosed and restored microwave and base station alarms across Nokia and Ericsson platforms (2G/3G/FD-LTE), ensuring high network availability.
- Performed predictive/preventive maintenance, site surveys, BTS parameter audits, and drive tests to optimize network performance.

Philippine EDS Techno-service Inc. (Yazaki Affiliate) June 2006 – February 2008
Junior Design Engineer / Team Lead | Philippines

- Designed wiring harness and system layouts for new automotive platforms (Japan & Europe markets) using HCAD, MS Visio, CATIA, and circuit simulation tools.
- Validated wire/fuse compatibility, performed circuit simulations, and coordinated design changes with international engineering teams.
- Served as Team Lead and quality checker for 2D/3D design outputs.

KEY PROJECT

TicketFlow – Field Engineer Ticket & Restoration Management System (2026)
Full-stack web application developed with Generative AI and Agentic AI assistance to automate O&M ticket management for field engineers. Features include OCR-based ticket creation from photos, SLA tracking (24h/48h/72h) with color-coded status, automated Restoration Reports with evidence photos and digital signatures, real-time dashboard, and mobile access.
Technologies: Python (Flask), HTML/CSS/JavaScript, OCR, SQLite, PDF generation.

EDUCATION

Master of Science in Electronics & Communications Engineering (Major in Control Systems)
Mapúa Institute of Technology, Philippines • Completed 31 graduate units • Emerson Process Management (EPM) Company Scholar

Bachelor of Science in Electronics & Communications Engineering
Southern Luzon State University, Philippines • DOST-SEI Scholar

LICENSES & CERTIFICATIONS

- Registered Engineer (E) — Saudi Council of Engineers (SCE No. 241763)
- Conferred ASEAN Engineer (AE) — ASEAN Federation of Engineering Organisations (AFEO No. 7542)
- Licensed Professional Electronics Engineer (PECE) — PRC, Philippines (Lic. No. 0000992)
- Licensed Electronics Engineer (ECE) — PRC, Philippines (Lic. No. 0036118)
- Certified General Security Professional (CGSP) – CCTV | Hikvision • Valid: Dec 2025 – Dec 2030

TECHNICAL SKILLS

Programming & Analytics: Python, PowerShell, Power BI, Generative AI & Agentic AI-assisted development, Flask, SQLite
Telecom & OSS: Ericsson (OSS-RC/WinFIOL/OMT), Huawei BTS Site Maintenance Terminal, Nokia (MMI/BTS Manager), IDEMIA Fusion Manager
Transmission Tools: AVIAT Portal, Nokia Hopper Manager, Alcatel AWY 9400, LCT-XDM
Test Equipment: Anritsu Site Master / OTDR / Power Meter, Fluke Multitester, EXFO Testers
Operating Systems: Windows, Linux, macOS